

When is the Gram matrix of MGD unstable, and what we do about it

A step-by-step explanation, with the measurements of the 30 September run

1 October 2026

What this note is for

At every time step, MGD moves the walkers by solving a linear system whose matrix is the *Gram matrix* G of the gradients of the statistics. If G is “unstable” (the word used in numerics is *ill-conditioned*), the solution can become huge or meaningless. This note explains, without assuming any background in numerical linear algebra:

1. where G comes from (Sections 1–2);
2. the two ways in which G can make the solve fail (Section 3);
3. the three remedies used in the code: rescaling, ridge, mask (Section 4);
4. what we measured along the transport in the run of 30 September (Section 5);
5. what this does and does not explain (Section 6).

[measured] marks a number computed on the run or in a controlled test; [hypothesis] marks an interpretation that has not been tested directly.

1 MGD in one page

Statistics. A signal is a vector $x \in \mathbb{R}^M$ (here $M = 256$). We describe it by r real-valued *statistics* $\phi_1(x), \dots, \phi_r(x)$ (here $r = 281$): L^6 norms of wavelet coefficients, region statistics, scattering coefficients, We write $\phi = (\phi_1, \dots, \phi_r)$. Each ϕ_n has a gradient $\nabla \phi_n(x) \in \mathbb{R}^M$: the direction in which one should move the signal x to increase ϕ_n fastest.

Walkers and their moments. MGD evolves N signals $x^1, \dots, x^N$, the *walkers* (here $N = 8500$), from time $t = 0$ to $t = 1$ on a grid $t_0 < t_1 < \dots < t_{n_t}$. The walkers at step k are written x_k^i . Their *empirical moments* are the averages

$$\bar{\phi}_n(x_k) = \frac{1}{N} \sum_{i=1}^N \phi_n(x_k^i), \quad n = 1, \dots, r.$$

Target moments. Let $x_1^1, \dots, x_1^P$ be the data and $x_0^1, \dots, x_0^P$ independent Gaussian noise. The *interpolant* between noise and data is

$$I_t^j = \cos\left(\frac{\pi t}{2}\right) x_0^j + \sin\left(\frac{\pi t}{2}\right) x_1^j, \quad j = 1, \dots, P,$$

pure noise at $t = 0$, the data at $t = 1$. The *target moments* at time t are

$$m_n^*(t) = \frac{1}{P} \sum_{j=1}^P \phi_n(I_t^j).$$

No model of the data is needed: the targets are computed from data and noise. The aim of MGD is that, at every time, the walkers have the target moments, $\bar{\phi}_n(x_k) \approx m_n^*(t_k)$ for all n , while being otherwise as random (maximum entropy) as possible.

One time step. With $h = t_{k+1} - t_k$, each step has two parts (code: `iteration_step_projection`).

- **Predictor.** Move the walkers along the gradients so that their moments follow the motion of the target, and add noise of strength σ (here $\sigma = 3.5$):

$$y_k^i = x_k^i + h \sum_{m=1}^r \eta_m \nabla \phi_m(x_k^i) + \sqrt{2h} \sigma \xi^i, \quad \xi^i \sim \mathcal{N}(0, I).$$

- **Corrector.** Move the walkers once more along the gradients to remove the remaining gap to the target at t_{k+1} :

$$x_{k+1}^i = y_k^i + \sum_{m=1}^r \theta_m \nabla \phi_m(y_k^i). \quad (1)$$

The unknowns are the coefficients $\eta = (\eta_1, \dots, \eta_r)$ and $\theta = (\theta_1, \dots, \theta_r)$, one per statistic. (The θ saved to disk is divided by $h\sigma^2$; this does not matter here.) Both are obtained by solving a linear system with the Gram matrix. We now see why.

2 Where the Gram matrix comes from

Take the corrector (1). How much does it change the moment $\bar{\phi}_n$? For a small move, a function changes by (gradient)·(displacement); this is the first-order Taylor expansion:

$$\phi_n(x_{k+1}^i) \approx \phi_n(y_k^i) + \nabla \phi_n(y_k^i) \cdot (x_{k+1}^i - y_k^i) = \phi_n(y_k^i) + \sum_{m=1}^r \theta_m \nabla \phi_n(y_k^i) \cdot \nabla \phi_m(y_k^i).$$

Averaging over the walkers,

$$\bar{\phi}_n(x_{k+1}) - \bar{\phi}_n(y_k) \approx \sum_{m=1}^r G_{nm} \theta_m, \quad \boxed{G_{nm} = \frac{1}{N} \sum_{i=1}^N \nabla \phi_n(y_k^i) \cdot \nabla \phi_m(y_k^i)}$$

G is the **Gram matrix of the gradients**: entry (n, m) is the inner product of the gradients of statistics n and m , averaged over the walkers. We want the change to close the gap to the target, $b_n = m_n^*(t_{k+1}) - \bar{\phi}_n(y_k)$. This gives one linear equation per statistic,

$$G \theta = b. \quad (2)$$

The predictor gives the same kind of system, $G \eta = \dot{m}^*(t_k)$, with G evaluated on x_k and the right-hand side equal to the speed at which the target moments move.

Reading G .

- The **diagonal entry** $G_{nn} = \frac{1}{N} \sum_i |\nabla \phi_n(y_k^i)|^2$ says how strongly statistic n reacts when the walkers move. If $\nabla \phi_n = 0$ on every walker, no move of the form (1) can change $\bar{\phi}_n$.
- The **off-diagonal entry** G_{nm} says how much moving along $\nabla \phi_m$ also changes ϕ_n . If the two gradients are orthogonal on average over the walkers, $G_{nm} = 0$: the two statistics can be adjusted independently. If they are almost parallel, adjusting one also adjusts the other.

3 Two ways in which the solve can fail

Solving (2) means finding the coefficients θ that produce the requested changes b . It fails when some requested change can only be produced by an enormous move. This happens in two distinct ways.

3.1 Failure 1: a statistic that the walkers cannot move

Take the simplest case: the gradient of statistic n is orthogonal, on average over the walkers, to the gradients of all the others ($G_{nm} = 0$ for every $m \neq n$). Row n of (2) then reads $G_{nn}\theta_n = b_n$, so

$$\theta_n = \frac{b_n}{G_{nn}}.$$

The size of the resulting move is $\frac{1}{N} \sum_i |\theta_n \nabla \phi_n(y_k^i)|^2 = \theta_n^2 G_{nn} = b_n^2 / G_{nn}$. If $G_{nn} \rightarrow 0$ and the gap b_n does not shrink at least as fast as $\sqrt{G_{nn}}$, the move diverges: we ask for a change of a moment that the walkers can barely influence. Nothing in the method forces b_n to shrink that fast: b_n is the difference of two averages over finitely many samples, and their sampling noise does not depend on G_{nn} .

Where it happens. Region statistics (`Scalar*_gaussianK`) measure how much of a wavelet channel falls in a given range of amplitudes. Early in the transport the walkers are close to Gaussian noise, and some amplitude ranges contain almost no walker. There $\nabla \phi_n \approx 0$ and $G_{nn} \approx 0$ (on the walkers, G_{nn} reached 10^{-16} – 10^{-8} times its data value [\[measured\]](#)). The walker blow-up of job 268295 at step 163 (28 September) was diagnosed as this, and the rerun with the mask of Section 4 completed [\[measured\]](#).

3.2 Failure 2: two statistics that move the walkers in the same way

Now take two statistics a and b whose gradients are almost the same (after normalising both to the same size). With unit diagonal, their 2×2 block of G is

$$G = \begin{pmatrix} 1 & 1 - \delta \\ 1 - \delta & 1 \end{pmatrix}, \quad \delta \text{ small.}$$

Solving $G\theta = b$ by hand (add and subtract the two equations):

$$\theta_a + \theta_b = \frac{b_a + b_b}{2 - \delta}, \quad \theta_a - \theta_b = \frac{b_a - b_b}{\delta}.$$

The *sum* $\theta_a + \theta_b$ is well determined. The *difference* is divided by δ . Both statistics push the walkers in (almost) the same direction, so the walkers only “see” their sum, and the difference is decided by a tiny δ . Whatever is in $b_a - b_b$, including sampling noise (targets and walker moments are averages over finitely many samples) and rounding errors, is multiplied by $1/\delta$. For instance, with $\delta = 10^{-7}$, a difference $b_a - b_b = 10^{-6}$ gives $\theta_a - \theta_b = 10$. Whether the result is actually large depends on how small $b_a - b_b$ happens to be; the point is that the solve has no control over it.

Note that *neither statistic is weak*: both have $G_{aa} = G_{bb} = 1$. Failure 2 is invisible on the diagonal; it is in the off-diagonal entry $1 - \delta$.

3.3 The general picture: eigen-directions and the condition number

Both failures are special cases of one picture. G is symmetric, so it has r orthogonal *eigen-directions* $u_1, \dots, u_r$ (each is a combination of the r coefficients) with eigenvalues $\mu_1 \leq \mu_2 \leq \dots \leq \mu_r$, all ≥ 0 . The solution of (2) is

$$\theta = \sum_{i=1}^r \frac{u_i \cdot b}{\mu_i} u_i.$$

Read it as: the eigenvalue μ_i says how much the moments change when the coefficients move along u_i ; to obtain the requested change, the solve divides by μ_i . A **small eigenvalue** is a combination of coefficients that the walkers barely see, and whatever part of b lies along it is blown up. In the 2×2 example, the eigenvalues are $2 - \delta$ (direction “ $a + b$ ”) and δ (direction “ $a - b$ ”). In Failure 1, the small eigenvalue is G_{nn} itself, with direction “statistic n alone”.

The **condition number** is the ratio of the largest to the smallest eigenvalue,

$$\kappa = \frac{\mu_{\max}}{\mu_{\min}}.$$

Rule of thumb: errors in G or b can be amplified by up to a factor κ in the solution, i.e. one can lose about $\log_{10} \kappa$ significant digits. A computer number in single precision (`float32`) carries about 7 digits, in double precision (`float64`) about 16. So when κ approaches 10^7 , a matrix stored in `float32` no longer determines the solution along its weakest directions.

4 The three remedies in the code

1. Rescaling (Jacobi). The statistics have different units, so the raw G mixes large and small numbers for no physical reason. The code first divides each statistic by $\sqrt{G_{nn}}$:

$$\tilde{G}_{nm} = \frac{G_{nm}}{\sqrt{G_{nn}G_{mm}}}, \quad \tilde{G}_{nn} = 1.$$

$\tilde{G}_{nm}$ is then the *cosine* between the two gradients, each seen as one long vector over all walkers and all positions: 0 if they are orthogonal on average, close to ± 1 if almost parallel. All condition numbers below are for $\tilde{G}$. Rescaling removes differences of units but cannot fix either failure: in Failure 2 the cosine stays $1 - \delta$, and in Failure 1 the division is by the vanishing $\sqrt{G_{nn}}$ itself.

2. Ridge ε . The code solves $(\tilde{G} + \varepsilon I)\tilde{\theta} = \tilde{b}$ with $\varepsilon = 10^{-4}$ (**--regularization**). In the eigen-picture,

$$\tilde{\theta} = \sum_i \frac{u_i \cdot \tilde{b}}{\mu_i + \varepsilon} u_i :$$

directions with $\mu_i \gg \varepsilon$ are untouched; directions with $\mu_i \ll \varepsilon$ get the coefficient $(u_i \cdot \tilde{b})/\varepsilon$ instead of $(u_i \cdot \tilde{b})/\mu_i$. So the ridge *caps* Failure 2, at the price that the coefficient along these directions is set by the arbitrary ε and not by the data.

The ridge does *not* cure Failure 1. Undoing the rescaling, the solved system is $(G + \varepsilon \text{diag } G)\theta = b$. In the uncoupled case this gives $\theta_n = b_n/((1 + \varepsilon)G_{nn})$: the ridge is proportional to G_{nn} and vanishes with it.

3. The mask (for Failure 1). At each step we keep only the statistics that can still act on the walkers,

$$L = \{ n : G_{nn} > \tau_n \},$$

with $\tau_n = 10^{-6}$ for the region statistics and $\tau_n = 0$ for all others. Each statistic is normalised so that $\mathbb{E}_{\text{data}} |\nabla \phi_n|^2 = 1$, so $\tau_n = 10^{-6}$ means one millionth of its value on the data. In practice, row and column n of G are removed for every $n \notin L$: we solve the smaller system $(G_{LL} + \varepsilon \text{diag } G_{LL})\theta_L = b_L$ and set $\theta_n = 0$ for the others. This barely affects the live statistics, because by the Cauchy–Schwarz inequality $|G_{nm}| \leq \sqrt{G_{nn}G_{mm}}$: when G_{nn} is close to zero, the whole row and column n are close to zero as well. Setting $\theta_n = 0$ removes almost no control, since the walkers could barely move $\bar{\phi}_n$ anyway; the price is that $\bar{\phi}_n$ is not corrected while the statistic is masked. When the amplitude range fills up again, $G_{nn} > \tau_n$ and the statistic re-enters L automatically.

The mask looks only at the diagonal, so it cannot see Failure 2: two near-copies each have a normal diagonal.

5 What we measured along the transport (run of 30 September)

Job 382722: turbulence, seed 900, two-phase time grid ($n_t = 60\,000$), $N = 8500$, $r = 281$, $\varepsilon = 10^{-4}$. The run logged the eigenvalues of $\tilde{G}$ every 10 steps (6000 logged steps), for both solves; the figures are in `turbulence/gram_along_transport.ipynb`. The run finished normally [measured].

5.1 Failure 1: masking

Only 5 of the 281 statistics were ever masked, all region statistics, all before $t = 0.15$ [measured]:

statistic	masked during	masked steps
Scalar_morlet_gaussianK[21]	$t < 0.021$	236
Scalar_psi_gaussianK[55]	$t < 0.044$ (short breaks at $t \approx 0.021$ and 0.0435)	478
Scalar_psi_gaussianK[33]	$t < 0.055$ (short break at $t \approx 0.021$)	603
Scalar_morlet_gaussianK[30]	$t < 0.148$	1646
Scalar_psi_gaussianK[77]	$t < 0.149$	1657

Seven statistics have a coefficient jump, defined as $|\theta_n|$ above 100 times its own running median. For all seven, the largest jump is at $t \approx 0.0213$ – 0.0215 [measured]. This is the moment when `Scalar_morlet_gaussianK[21]` is switched back on, with little margin above its threshold, and `Scalar_psi_gaussianK[33]` and `[55]` are briefly switched on as well [measured]. That the re-activation causes the jumps is likely but not proven [hypothesis]. Four of the seven statistics have a second, smaller jump episode, whose time I have not located.

5.2 Failure 2: near-copies

The four smallest eigenvalues of $\tilde{G}$ (corrector), median over each time window [measured]:

t window	μ_1	μ_2	μ_3	μ_4	$\kappa = \mu_{\max}/\mu_1$
[0, 0.1)	$6.0 \cdot 10^{-7}$	$4.4 \cdot 10^{-6}$	$2.6 \cdot 10^{-5}$	$2.7 \cdot 10^{-4}$	$2.5 \cdot 10^7$
[0.1, 0.5)	$1.9 \cdot 10^{-7}$	$5.0 \cdot 10^{-7}$	$5.3 \cdot 10^{-6}$	$3.5 \cdot 10^{-4}$	$6.2 \cdot 10^7$
[0.5, 0.9)	$1.6 \cdot 10^{-7}$	$6.7 \cdot 10^{-7}$	$1.2 \cdot 10^{-5}$	$5.1 \cdot 10^{-4}$	$7.2 \cdot 10^7$
[0.99, 0.999)	$1.5 \cdot 10^{-7}$	$6.7 \cdot 10^{-7}$	$1.3 \cdot 10^{-5}$	$5.1 \cdot 10^{-4}$	$1.3 \cdot 10^8$
[0.999, 1)	$1.4 \cdot 10^{-7}$	$6.7 \cdot 10^{-7}$	$1.3 \cdot 10^{-5}$	$5.0 \cdot 10^{-4}$	$1.5 \cdot 10^8$

Three eigenvalues are below the ridge $\varepsilon = 10^{-4}$ at almost every logged step (two or four at a few steps near $t = 0$), and μ_4 is a factor 10–70 above μ_3 [measured]. So $\tilde{G}$ has **three directions far weaker than all the others**.

Which statistics. For the smallest direction, the log records the statistics with the largest weight. At 96% of the steps it is the pair $L_6[7]$ / $L_6^\psi[20]$; at 3% the pair $L_6[8]$ / $L_6^\psi[21]$; at a few steps the pair `Scalar_psi_gaussianK[77]` / `Scalar_morlet_gaussianK[30]` [measured]. A search for numerically null directions on the saved systems (25 times checked) finds, at $t = 0.9985$, the direction $0.71 [L_6[7]] - 0.71 [L_6^\psi[20]]$: exactly the “ $a - b$ ” direction of Section 3.2 [measured]. The log records the composition of the smallest direction only. That μ_1 and μ_2 are the two L_6 pairs is therefore an inference from which pairs show up as smallest, and the third direction (μ_3) is probably the Scalar pair; neither has been checked directly [hypothesis].

Why these pairs exist. L_6 is computed on a wavelet bank with $Q = 1$, L_6^ψ on a bank with $Q = 3$. In the usual construction of such banks, Q sets the bandwidth *relative* to the centre frequency. But the grid has $M = 256$ points, so frequencies only come in integer bins. At the coarsest scales, both filters of each pair put essentially all their weight on the same one or two

bins (bin 1 for the first pair, bin 0 for the second) [measured], so they are almost identical. Measured as $1 - |\cos|$ between the filters, they differ by $5.8 \cdot 10^{-4}$ and $1.5 \cdot 10^{-3}$; between the gradients of the corresponding L_6 statistics, on synthetic signals and in `float64`, the difference is only $4 \cdot 10^{-7}$ – $1.4 \cdot 10^{-5}$ [measured]. The code already removes duplicate filters *inside* the $Q = 3$ bank, but never compares the two banks with each other.

How reliable are these numbers. G is accumulated in `float32` (about 7 digits) and only then converted to `float64` for the solve. Rounding each entry to 7 digits moves the eigenvalues by an amount that, for a 281×281 matrix, can reach roughly 10^{-7} – 10^{-6} [hypothesis] (order-of-magnitude estimate, not measured). At 5% of the logged steps the smallest eigenvalue indeed comes out slightly *negative* ($\approx -10^{-7}$), which is impossible for a Gram matrix [measured]. So:

- μ_1 and μ_2 ($\sim 10^{-7}$ – 10^{-6}) are at the precision limit. Their order of magnitude is consistent with the `float64` test on synthetic signals above, but their values, and hence $\kappa \sim 10^7$ – 10^8 , cannot be trusted precisely. What is certain is that these directions are far below the ridge.
- μ_3 and μ_4 ($\sim 10^{-5}$ and $\sim 5 \cdot 10^{-4}$) are well above the precision limit.

Only building G in `float64` would give the true μ_1, μ_2 .

5.3 Without the near-copies

Removing k statistics from G (deleting their rows and columns) cannot make the new smallest eigenvalue larger than the old μ_{k+1} ; this is a standard result, *eigenvalue interlacing*. The bound is approximately reached when the removed statistics are the ones that make up the weak directions. The largest eigenvalue barely changes when 2–3 of 281 statistics are removed. So the table above gives, approximately, the best condition number that removing statistics can achieve:

statistics removed	new smallest eigenvalue	best κ
none	$\mu_1 \approx 1.5 \cdot 10^{-7}$	$\sim 10^8$ (at the precision limit)
2 statistics (one per L_6 pair)*	$\mu_3 \approx 1.2 \cdot 10^{-5}$	$\sim 10^6$
3 statistics (also the third direction)	$\mu_4 \approx 5 \cdot 10^{-4}$	$\sim 2\text{--}5 \cdot 10^4$

* assuming μ_1, μ_2 are the two L_6 pairs (see above).

The last value stays between $2 \cdot 10^4$ and $5 \cdot 10^4$ in every time window, from $t = 0$ to $t = 1$ [measured]. Near $t = 1$ the walkers should be distributed approximately like the data, so this suggests that $\sim 3 \cdot 10^4$ is also the conditioning of the full set of 281 statistics on the data, i.e. a property of the statistics rather than something the transport creates [hypothesis]. A direct check is to compute $\tilde{G}$ on the data. At that level the ridge still acts, but mildly: along the weakest remaining direction the solution is multiplied by $\mu_4/(\mu_4 + \varepsilon) \approx 0.7$ – 0.8 , and by almost 1 along all the others. `float32` has a comfortable margin.

Correction to the note of 30 September. The value $\kappa \approx 12$ quoted there was for the 73 `Scalar_psi_gaussianK` statistics alone, on the data. The full 281-statistic Gram contains near-dependencies *between* families (e.g. L_6 on $Q = 1$ and on $Q = 3$ filters) that a single family cannot show.

6 What this explains, and what it does not

- **Explained.** The crash of 28 September (Failure 1, cured by the mask), and why $\tilde{G}$ has a very large condition number (Failure 2: three weak directions, whose coefficients are set by ε ; the weakest certainly, and most likely the second, come from near-copy filters).

- **Not explained.** The moments that are still not matched at the end of the run: 11 moments miss their target by more than 1% at the last time, all of them region statistics, with errors that first exceed 1% between $t = 0.957$ and $t = 0.995$ [measured]. None of these statistics is ever masked, and none is ever among the main components of the smallest eigen-direction [measured]. We have no evidence linking them to the conditioning of G ; but the composition of the second and third weakest directions is unknown, so this is not excluded. Their cause is under investigation.

About the supervisor’s suggestion (conjugate gradient with early stopping). Stopping when $\|G\theta - b\|$ falls below the statistical noise of b is another way of regularising, in which the noise, not a hand-picked ε , sets the threshold (the *discrepancy principle*). Replayed on the logged systems, at 5999 of 6000 steps the gap b is already below its estimated noise level before any correction ($\theta = 0$), so the method would stop immediately and not correct at all [measured]. This holds with both the plain and the noise-weighted norm. The noise estimate itself may be too large, for instance if the noise of the target is not independent from step to step [hypothesis]; until that is checked, the criterion cannot be used as is.

Proposed next steps

1. Remove the cross-bank near-copies when the filters are built: drop the $Q = 1$ filters 7 and 8 (difference to a $Q = 3$ filter below 10^{-2} ; all other $Q = 1$ filters are at least $7 \cdot 10^{-2}$ away).
2. Identify the third near-singular direction, from the eigenvector of μ_3 on the saved systems.
3. Optionally, build G in `float64`, to measure the smallest eigenvalues beyond the `float32` limit.